

Limbic System

Principles and Practices of Osteopathic Medicine I

- *Jennifer Xie, Ph.D.*
- Associate Professor
- Biomedical and Anatomical Sciences
- Arkansas, Wilson Hall, Rm # 144J

Email: Jennifer.xie@nyit.edu



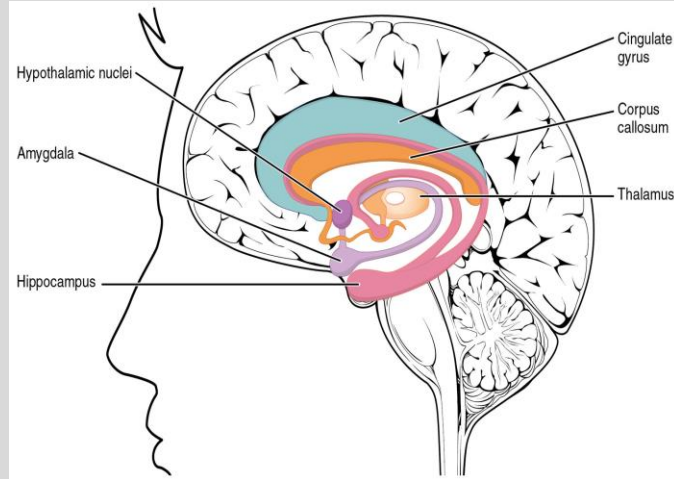
Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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1. Basic anatomy and cellular circuitry of the limbic system
2. Cingulate gyrus
3. Hippocampal formation and memory
4. Amygdala and emotion
5. Different kinds of memory disorders
6. Clinical syndromes associated with limbic system pathology



See **detailed objectives** in CPG
PLEASE READ THE Scholar Rx Brick!

Session Objectives Mapped to COMLEX Blueprint

- **Dimension 2**
 - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
 - Amnesia
 - Wernicke-Korsakoff encephalopathy
 - Learning disorders
 - Trauma and stressor-related disorders (eg, adjustment disorders, post-traumatic stress disorder)

Session Objectives Mapped to USMLE Blueprint

V. Nervous System & Special Senses

A. Normal Processes

- 2. Organ structure and function
 - iii. Brain
 - c. limbic system and emotional behavior circadian rhythm sleep-wake disorder

B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

- 2. Anxiety disorders
 - i – ix: acute stress disorder; anxiety disorders (generalized anxiety disorder, anxiety due to another medical condition, social phobia); hyperventilation syndrome; obsessive-compulsive disorder; panic disorder with and without agoraphobia; phobic disorders; post-traumatic stress disorder; separation anxiety disorder; substance-induced anxiety disorder, trichotillomania
- 3. Mood disorders
 - i – xi: major depressive disorder with and without psychotic features, with and without seasonal pattern; major depressive disorder, postpartum, with and without psychotic features, including screening; cyclothymic disorder; persistent depressive disorder (dysthymia); bipolar disorder, manic/depressed/mixed; premenstrual dysphoric disorder; bipolar and related disorder or depressive disorder due to another medical condition; substance/medication-induced bipolar and related disorder or depressive disorder (illegal or prescribed); suicidal ideation/attempt

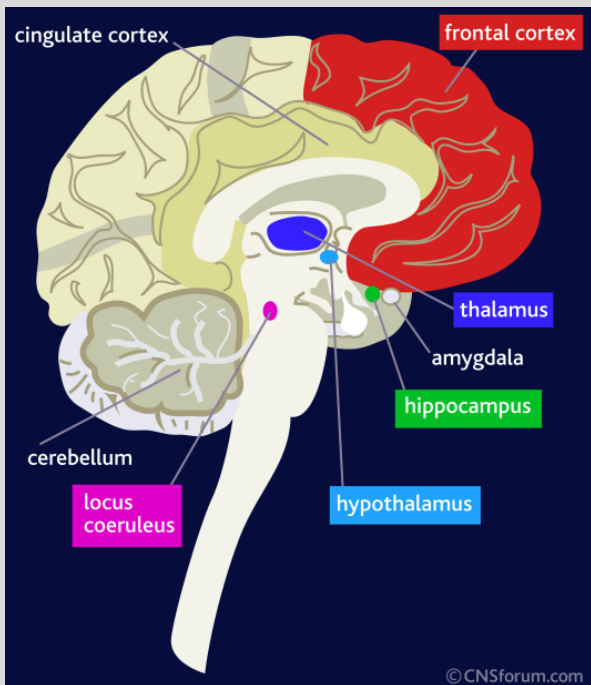
The Limbic System

Consists of a core of cortical and diencephalic structures to provide primitive **emotional** regulation of the **hypothalamic** and **brainstem autonomic** output.

Latin “Limbus” = “Border”:

- Bordering the cerebral cortex and diencephalon

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Composition under debate

Central components unanimously agreed on (the “**big 3**”):

- **Cingulate gyrus**
- **Hippocampal formation**
- **Amygdala**

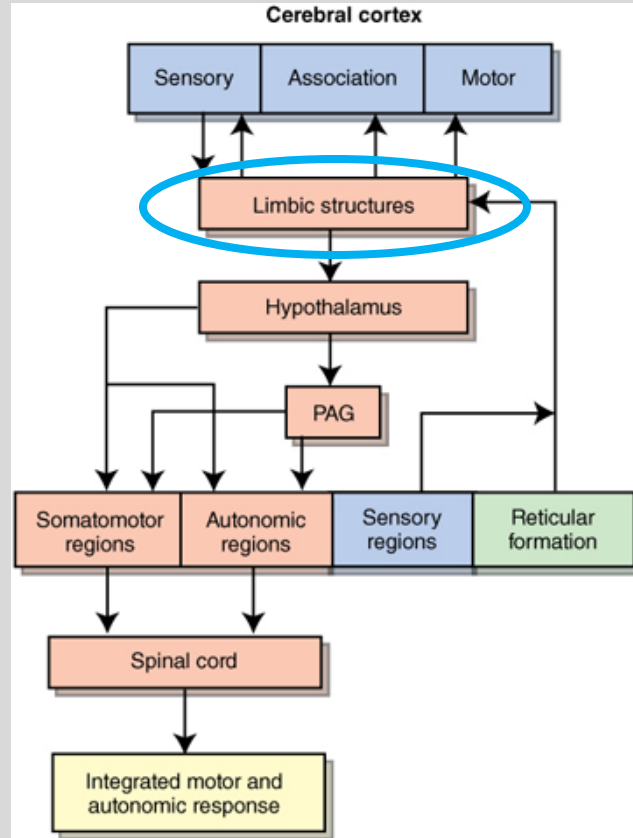
Other areas:

- **Entorhinal cortex**
- **Hypothalamus**
- **Mammillary body**
- **Anterior thalamic nuclei**
- **Dorsomedial nucleus of the thalamus**
- **Orbitofrontal cortex**
- **Septal area**
- **Prefrontal cortex**
- **Insula**

Connections of Limbic System

Receives **sensory information** from the cortex, spinal cord, and brainstem reticular formation.

Integrate the sensory information with emotional responses and send the primary output to **hypothalamus** to produce autonomic changes.



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Functions of the Limbic System

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The limbic system is involved in **primordial emotion**, **memory**, **olfaction**, attention, feeding and mating behavior. (**5 F's** - **F**eeding, **F**leeing, **F**ighting, **F**eeling, and **S**ex)

- Establishes emotional states
- Links conscious functions of cerebral cortex with autonomic functions of the brainstem
- Facilitates **memory** storage and retrieval



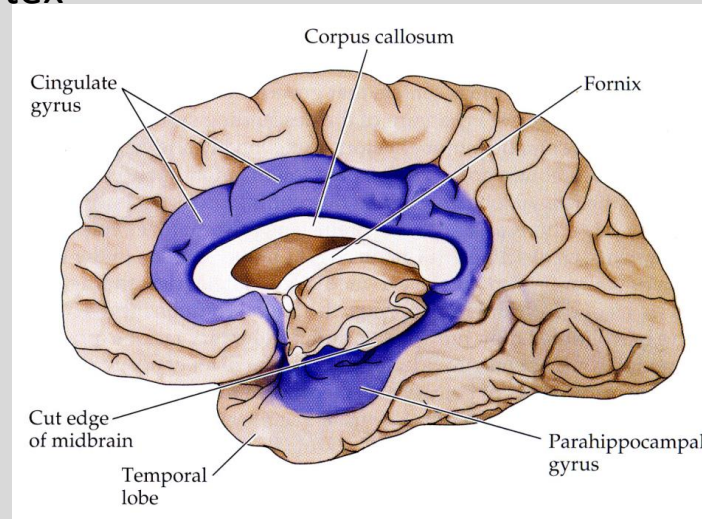
Limbic Lobe

Components of the limbic system:

- Limbic lobe of cerebral hemisphere
 - **Cingulate gyrus**
 - Parahippocampal gyrus
 - Medial orbitofrontal cortex

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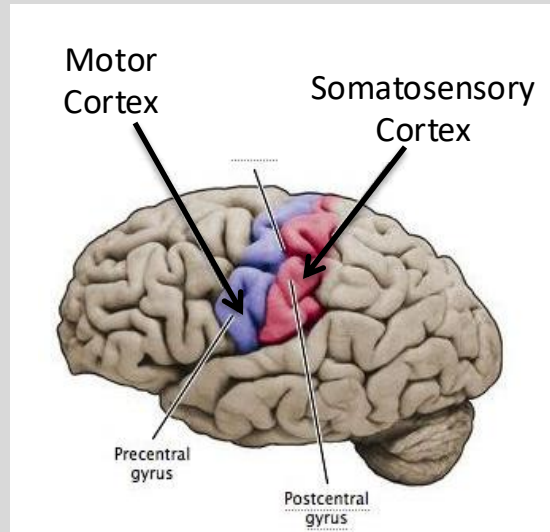
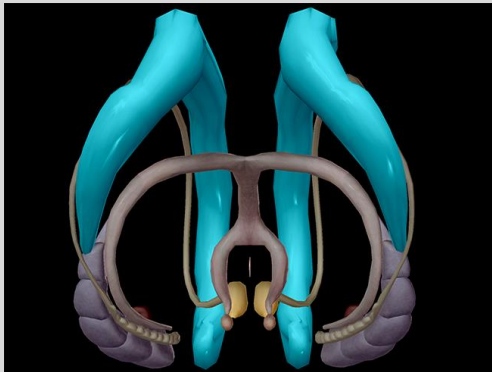
A.C. Brown, 2011

Cingulate Gyrus

- Gateway between cerebral cortex and limbic system
- Receives wide variety of input from the cortex and thalamus
- Projects to the cerebral cortex and hippocampus
- **Emotional** aspect of the **sensory perception** and **movement**

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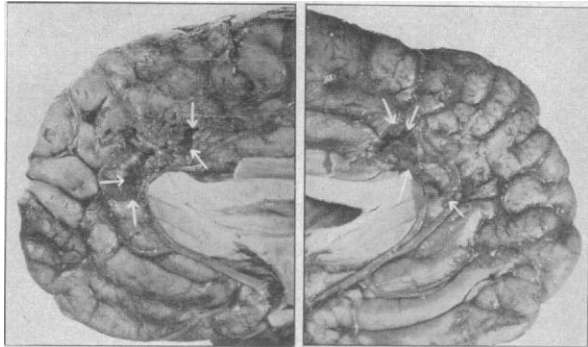
Cingulectomy for severe anxiety and depression

Section of Psychiatry with Section of Neurology

Effects of Anterior Cingulectomy in Man¹

By C. W. M. WHITTY, D.M., M.R.C.P.

Department of Neurology, Radcliffe Infirmary, Oxford



Improved psychiatric symptoms, but with reduced frontal lobe inhibition

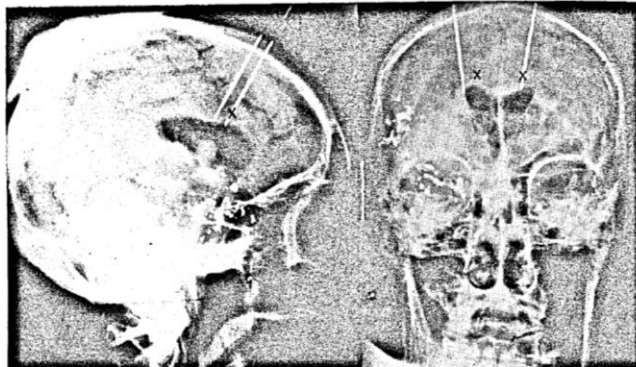
Cingulumotomy for Intractable Pain

PAIN “RELIEF” BY FRONTAL CINGULUMOTOMY*

ELDON L. FOLTZ, M.D., AND LOWELL E. WHITE, JR., M.D.

Division of Neurosurgery, University of Washington, Seattle, Washington

(Received for publication May 19, 1961)



“I can still feel pain”, but “it was not distressing”, “doesn’t bother me any more”

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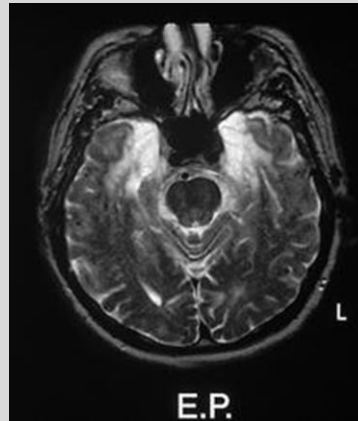
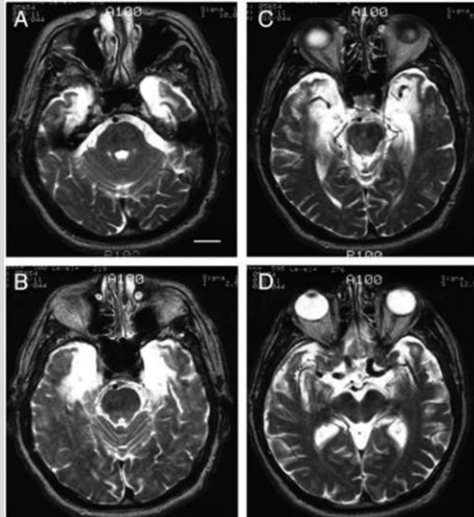
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Question

What symptoms do you think this patient experience?

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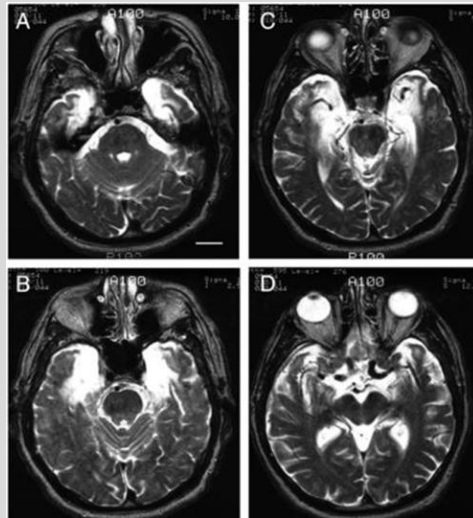
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Herpes Encephalitis

This patient became profoundly **amnesic** in 1992 after viral encephalitis, which damaged his medial temporal lobe bilaterally.

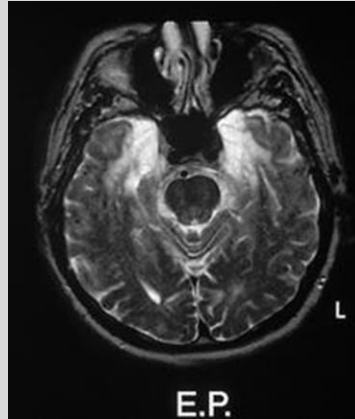
The **herpes simplex virus** has a tropism for the **limbic cortex** and can cause severe encephalitis involving primarily limbic cortical areas



Images are arranged from ventral (A) to dorsal (D), indicating the extent of damage in E. P.'s anterior temporal lobes (bright signal)

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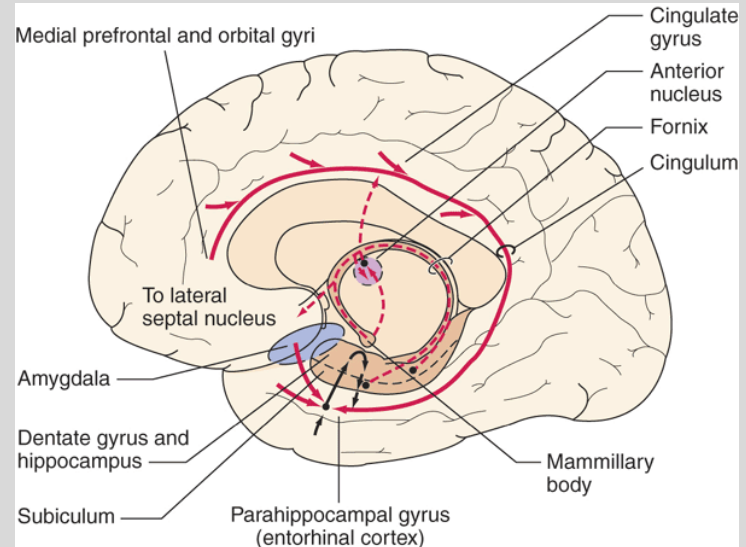
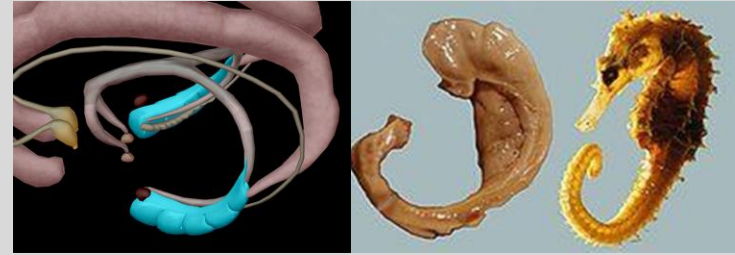
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Hippocampal Formation

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- Functions in **learning**, **memory** and **recognition of novel objects** and **navigation**.
- It receives major **input** through the **entorhinal cortex** and projects major **output** through the **fornix** to reach **diencephalon**.
- Composed of three major structures:
 - Dentate gyrus
 - Hippocampus
 - Subiculum



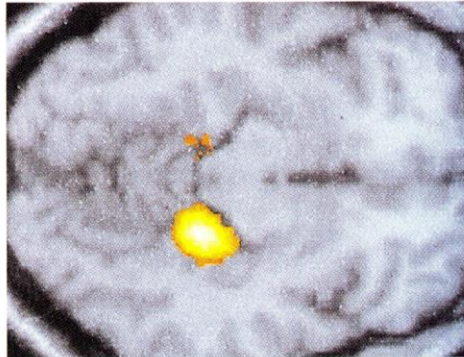
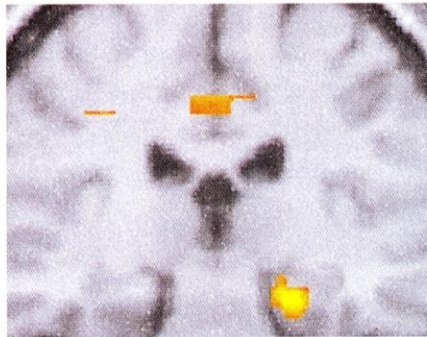
Right Hippocampus and Spatial Orientation

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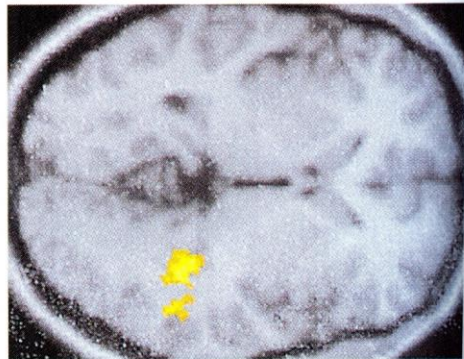
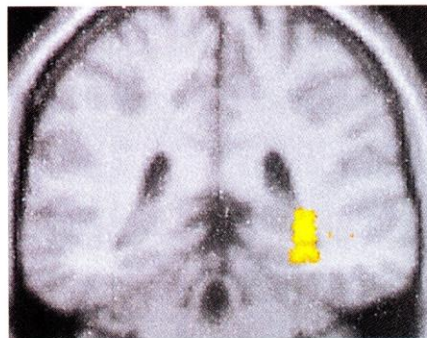
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Functional imaging of the brain while subjects watched a film that depicted navigation through the streets of an Irish town.

A Learning about surroundings (right hippocampus)



B Recall of taxi routes (right hippocampus)



L

R

Functional imaging of the brain of taxi drivers recalling routes around the city of London.

"I never noticed part of my brain growing - it makes you wonder what happened to the rest of it." David Cohen, taxi driver

(Kandel, E.R., Schwartz, J.H., Jessel, T.M.,
Principles of Neural Science, 4th ed.).

Intrinsic Organization of the Hippocampus

The hippocampus has four sectors/ areas: CA1, CA2, CA3, CA4 (CA = *Cornu Ammonis*)

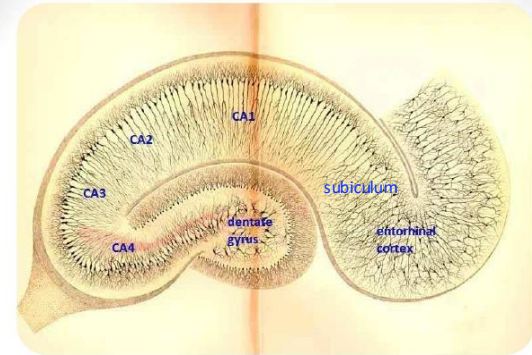
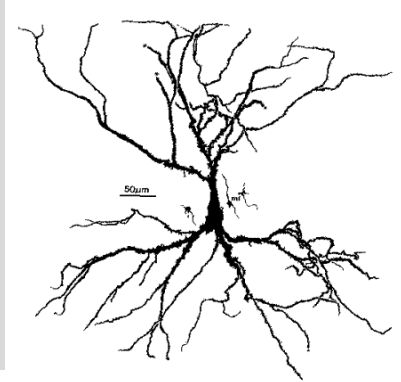
CA1 is adjacent to the subiculum and CA4 lies within the hilus of the dentate gyrus

Efferent fibers of the dentate gyrus synapse in CA3 pyramidal cells

Pyramidal cells in area **CA1** are exceptionally **sensitive to oxygen deprivation**

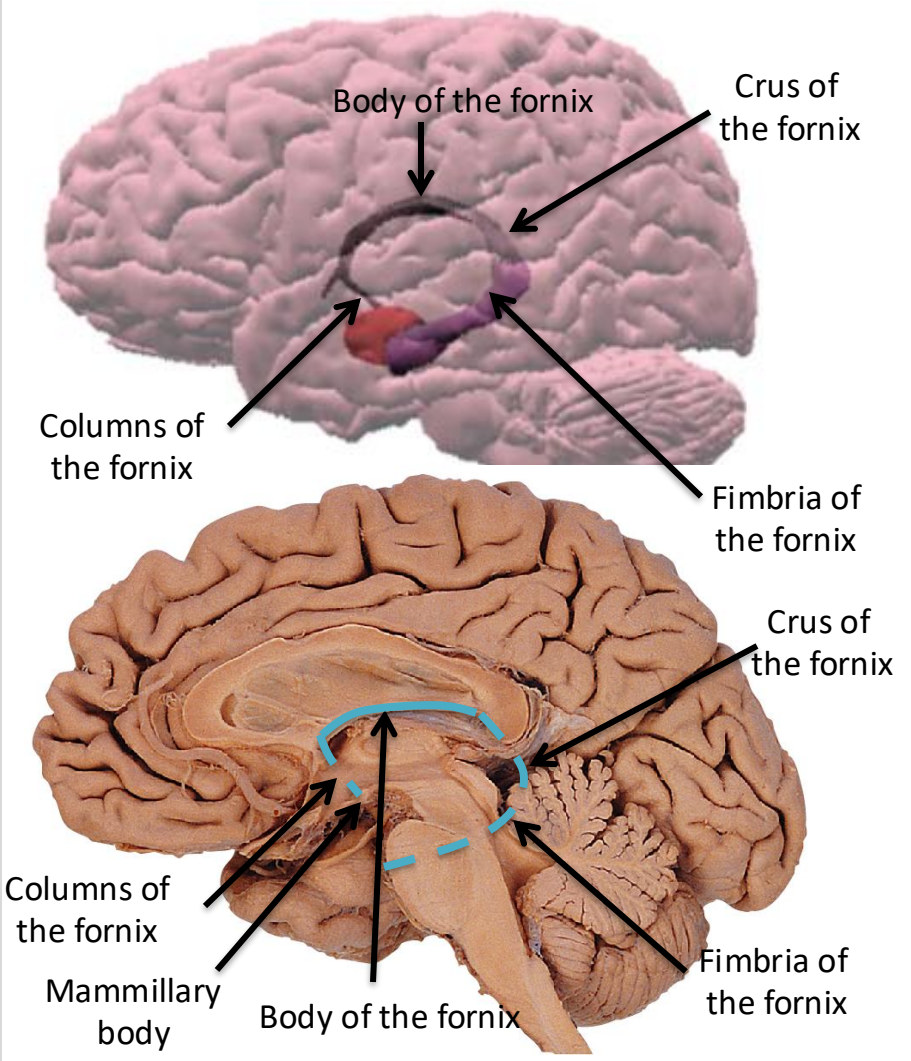
Cells die within minutes without a supply of fresh arterial blood

No need to remember the subdivisions – FYI only!!



Drawing of Hippocampus by Camilo Golgi
Subregions of the hippocampus exhibit histological differences.

Hippocampus & Fornix



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Hippocampus & Fornix

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Body

Hippocampus

Body

Fimbria

Crus

Anterior
Commissure

Columns

Temporal Lobe Epilepsy

Temporal Lobe Epilepsy

- The temporal lobe (especially the hippocampus and amygdala) has a **lower threshold for epileptic seizure** activity than the other cortical areas
- Seizures that originate in these regions are called **complex partial (psychomotor) seizures**
- Temporal lobe epilepsy may include **abnormal sensations, repeated involuntary movements** such as chewing, swallowing, and lip smacking, disorders of consciousness, **memory loss** and **hallucinations**

Amygdala

Almond shaped mass of nuclei located deep in the temporal lobes

It is a limbic structure that is involved in many of our **emotions** and **motivations**, particularly those that are related to **survival**

*It is important in processing of **emotions** such as **fear, anger and pleasure***

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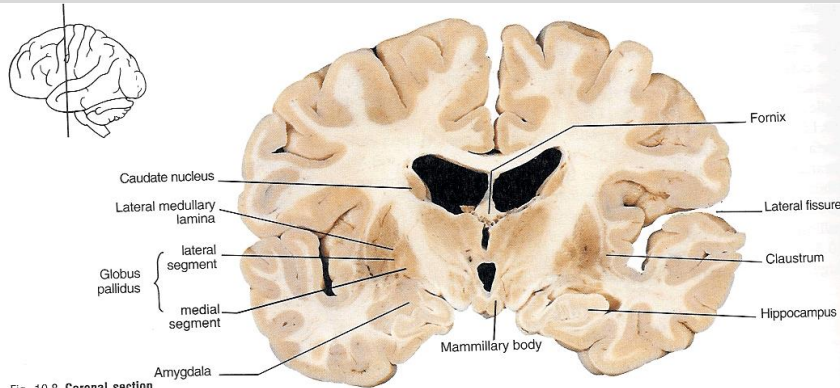
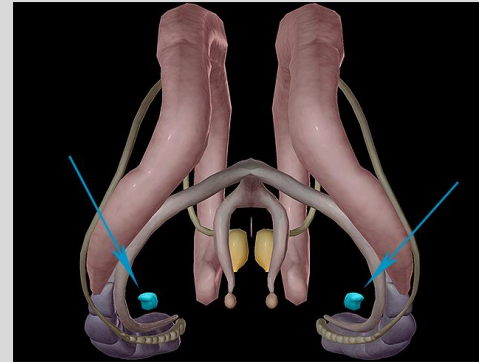


Fig. 10.8 Coronal section.

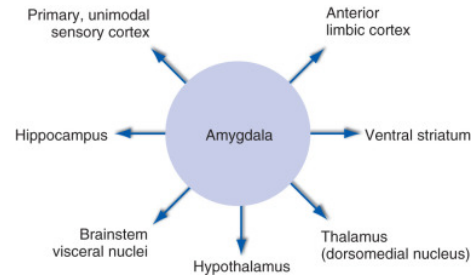
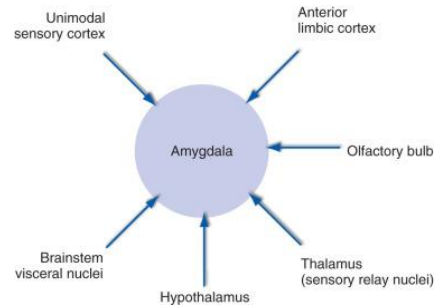
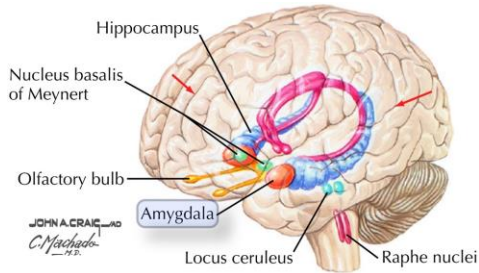


Amygdala and Emotions

Is a collection of about a dozen nuclei lying beneath the uncus of the limbic lobe, at the anterior end of the hippocampus and the inferior horn of the lateral ventricle

Output: to the **hypothalamus** and septal area

Input: sensory association cortices, hypothalamus, septal area, and hippocampal formation



Amygdala and Emotions

- Amygdala plays a pivotal role in **emotions** and **drives: Fear**
- Amygdala is an active participant in all major limbic functions
- **Reciprocal connections** between **Amygdala-Hypothalamus** mediate changes in heart rate, peristalsis, piloerection, sweating, and other changes seen with strong emotions
- It plays an important role in **attaching emotional significance to memories**
- **Olfactory tract projections to amygdala** may explain the occasional ability of odors to evoke emotions

Amygdala and Fear

Kluver-Bucy Syndrome

Bilateral lesions of the **amygdala**

These lesions result in:

Placidity: decrease in aggressive behavior

Psychic blindness: objects in visual field are treated inappropriately. Example: Monkeys approach snake with inappropriate docility

Hypermetamorphosis: visual stimuli are repeatedly approached as though they were completely new

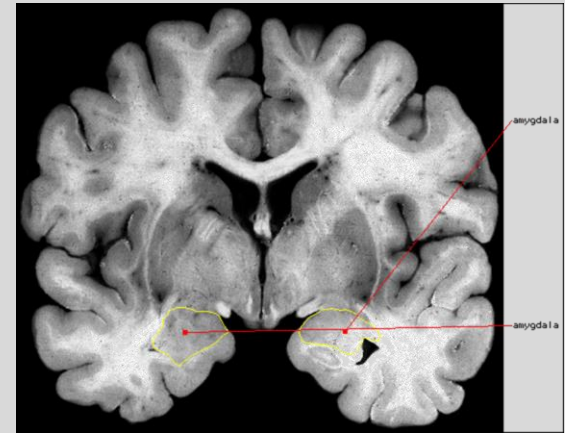
Hyper-orality: Increased oral exploratory behavior

Hypersexuality and loss of sexual preference

Anterograde amnesia: deficit in forming new memories

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Post-traumatic Stress Disorder (PTSD)

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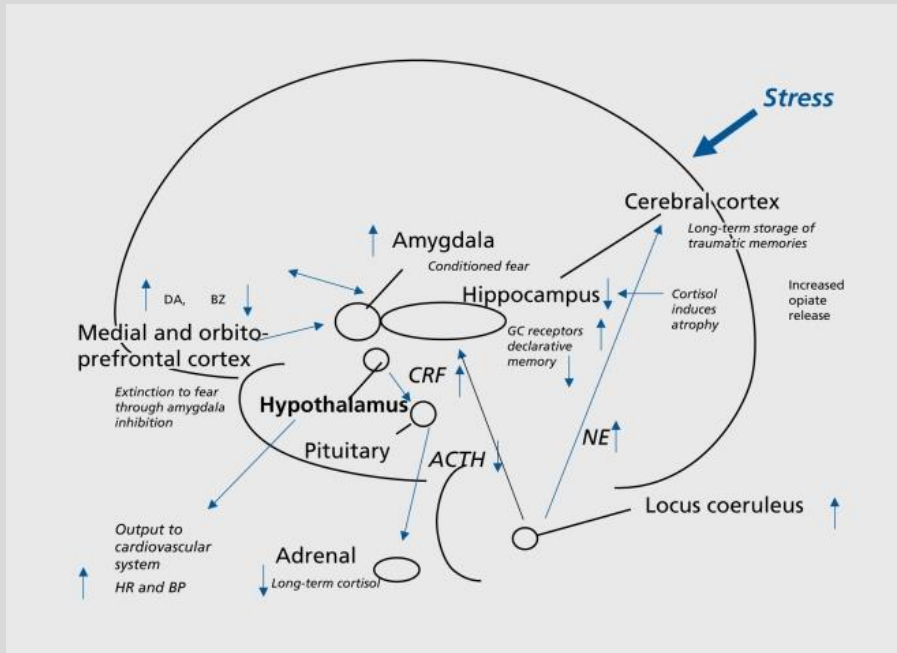


Figure 1. Lasting effects of trauma on the brain, showing long-term dysregulation of norepinephrine and Cortisol systems, and vulnerable areas of hippocampus, amygdala, and medial prefrontal cortex that are affected by trauma. GC, glucocorticoid; CRF, corticotropin-releasing factor; ACTH, adrenocorticotropin hormone; NE, norepinephrine; HR, heart rate; BP, blood pressure; DA, dopamine; BZ, benzodiazapine; GC, glucocorticoid

- PTSD is often characterized by **aberrant amygdala activation** and functional abnormalities in **cortico-limbic circuitry**
- Abnormalities in amygdala activity may be associated with **hyper-arousal** and **hypervigilance** to threat

Orbitofrontal Cortex

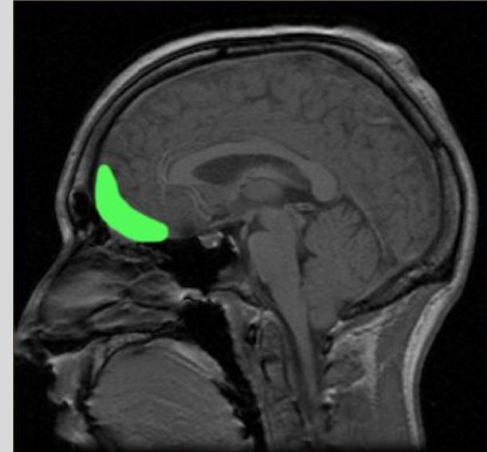
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It has reciprocal connections with the **dorsomedial nucleus** of the thalamus

It is connected through the medial forebrain bundle with the **septal area** and **hypothalamic** nuclei

*Primarily involved in **social/emotional processing** and conscious perception of **smell** (secondary olfactory cortex)*



Septal Area

It has reciprocal connections with the **hippocampal** formation through the fornix; and with the **hypothalamus** through the medial forebrain bundle

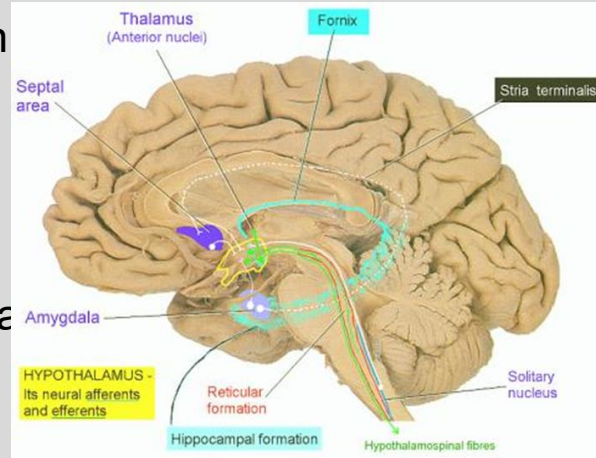
It serves as a **relay** for transmission of hippocampal impulses to the hypothalamus and as a feedback system to the hippocampal formation

In addition, activity in the septal area appears to be important in **reward** and **learning** (**dopamine mesolimbic** pathway)

- **Nucleus accumbens**
- Bed nucleus of the stria terminalis

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Hypothalamus

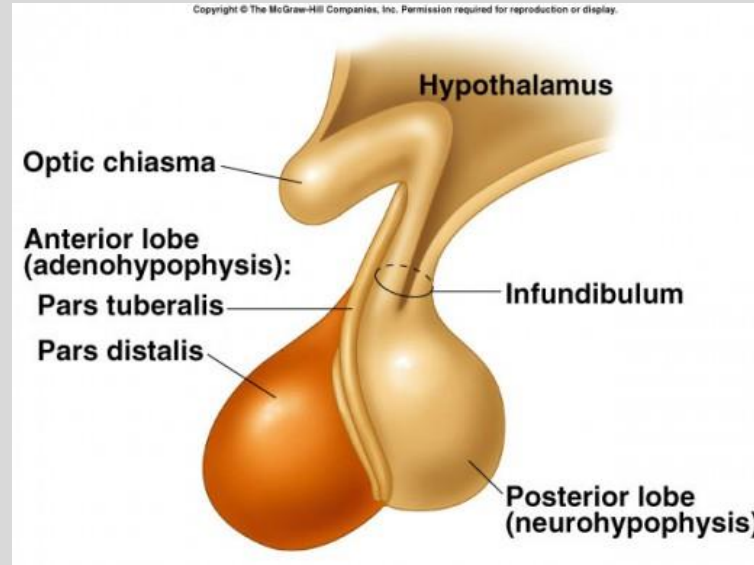
It has **reciprocal** connections with the **amygdala**. It influences various emotional responses

The hypothalamus is involved in several functions of the body including:

- **Autonomic Function Control**
- **Endocrine Function Control**
- **Homeostasis**
- **Motor Function Control**
- **Food and Water Intake Regulation**
- **Sleep-Wake Cycle Regulation**

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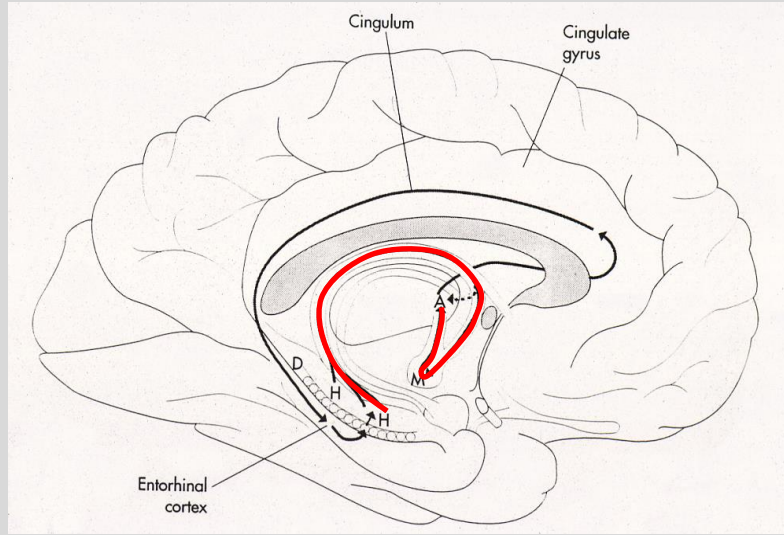
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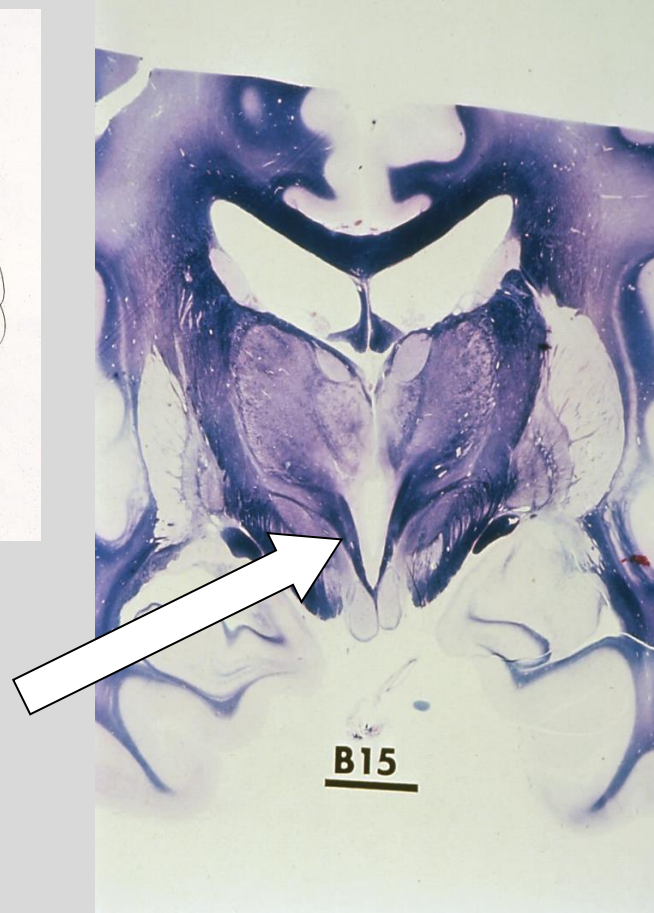
The mammillary bodies send information to the thalamus

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**via the
mammillothalamic tract**



Question

The Syndrome of Wernicke-Korsakoff is characterized by ataxia/nystagmus, ophthalmoplegia, and amnesia, confabulation, etc., respectively due to a lack of thiamine (Vitamin B1) from excessive alcohol consumption and malnutrition. The part of the CNS that is thought to be damaged/decreased activity is:

- A) the hippocampus
- B) the frontal cortex
- C) the parahippocampal gyrus
- D) the mammillary bodies**
- E) the hypothalamus



Wernicke-Korsakoff Syndrome

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Vitamin B1 deficiency in
alcoholics

Wernicke Encephalopathy

- Confusion
- Ataxia
- Nystagmus
- Ophthalmoplegia

“CAN Of beer” problem

Korsakoff Psychosis

- Amnesia (anterograde and retrograde)
- Hallucinations
- Confabulations

Treatment:

- **Thiamine** (Vitamin B1) 600 mg/day
- Reversible with early intervention
- Irreversible later on

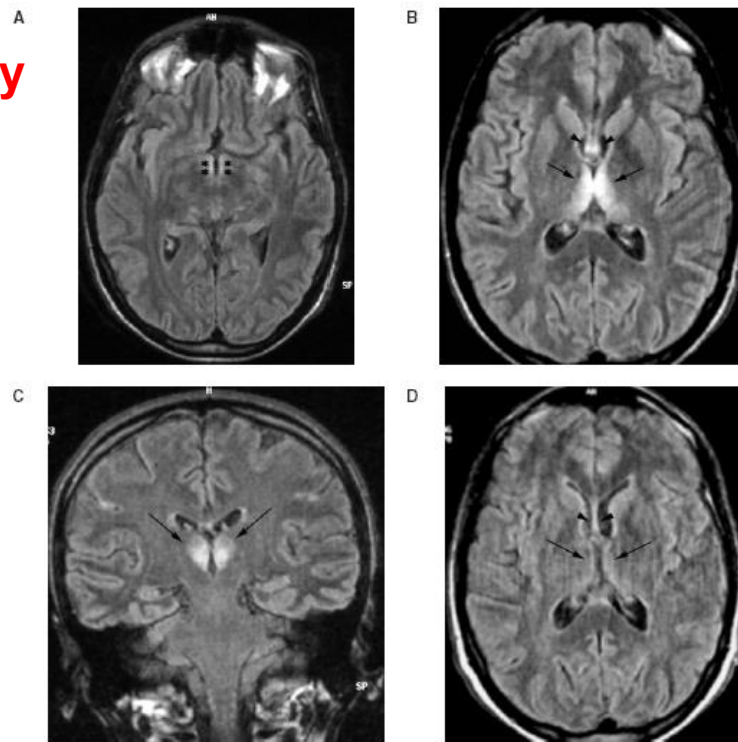
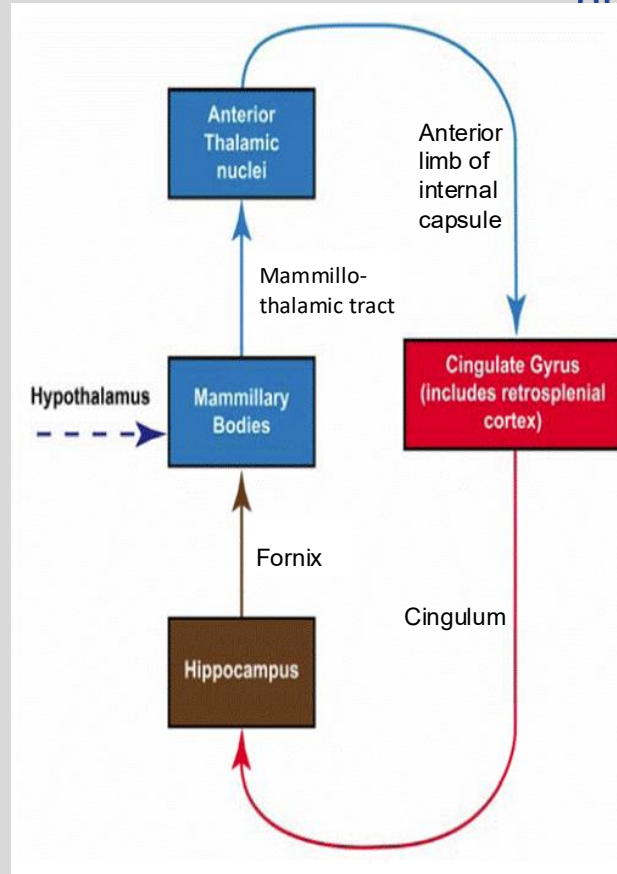


Fig. 2 Images from axial and coronal FLAIR sequence of the WK subject. Seven-day MRI study demonstrates signal abnormalities of the mammillary bodies (A) (stars), the medial aspect of the thalami (B, C) (arrows) and the fornix (B) (arrowheads). Five months later signal changes have almost completely disappeared (D) (arrows and arrowheads).

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Limbic Connections: The Papez Circuit

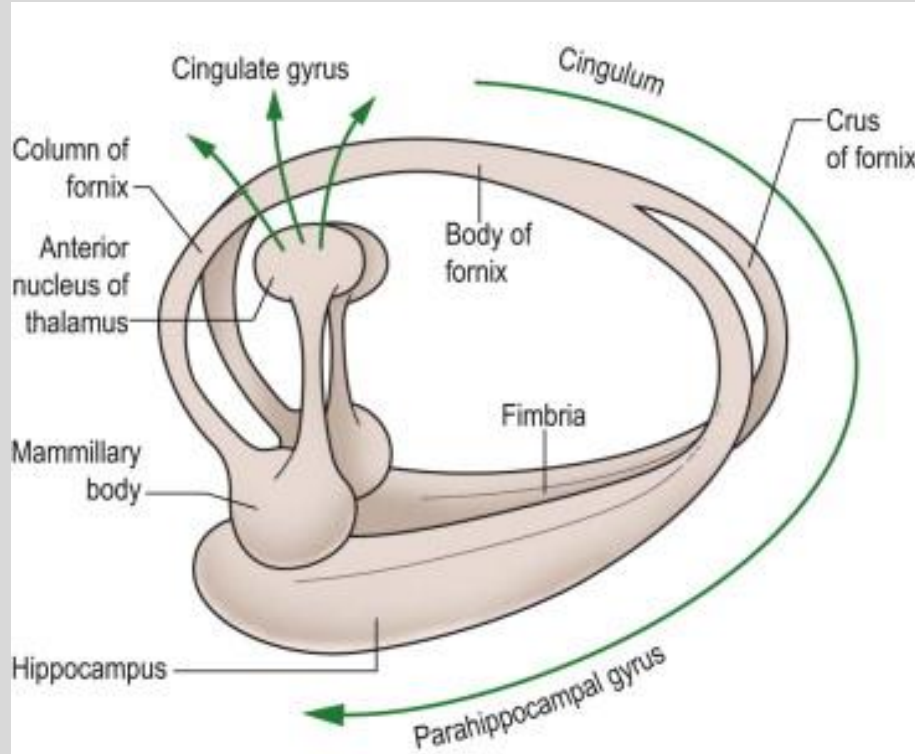
- Arbitrarily the Papez circuit begins and ends with the **hippocampus**
- Axons of the hippocampal pyramidal cells converge to form the fimbria and finally the **forix**, which projects to the **mammillary bodies** in the hypothalamus
- The mammillary bodies in turn project to the **anterior nucleus of the thalamus** by way of the **mammillothalamic tract**
- The anterior nuclei project to the **cingulate gyrus** through the **anterior limb of the internal capsule**
- The cingulate gyrus communicates with the **hippocampus** and **entorhinal cortex** through **cingulum**



As derived from JW Papez, 1937

Papez Circuit

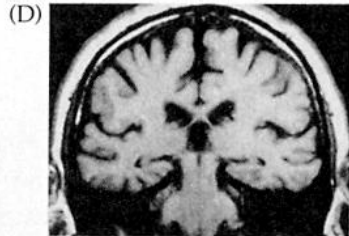
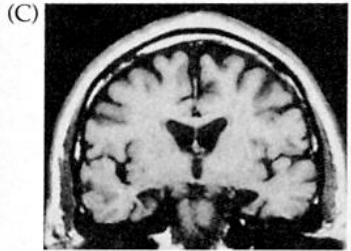
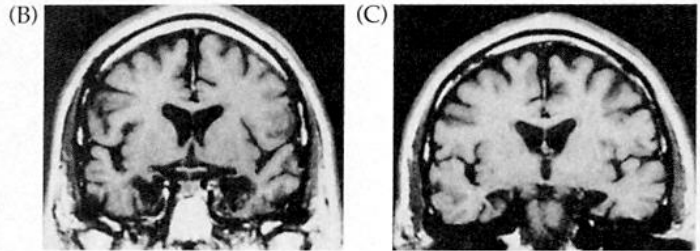
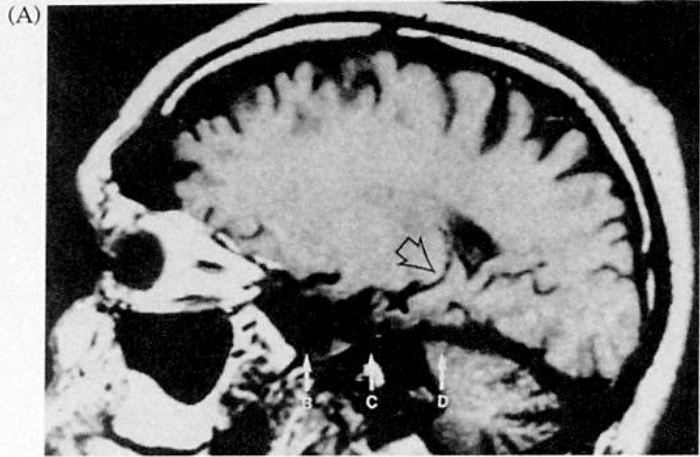
- Important for **episodic memory consolidation**
- Establish the **emotional significance** of memory



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Neuroanatomy: An Illustrated Colour Text, 5th ed., Alan Crossman and David Neary, 2015, Chapter 16



MRI of patient H.M. - much of the medial temporal lobes removed for treatment of intractable seizures.

H.M. had an inability to transfer short term memory to long term memory for events and facts.

([anterograde amnesia](#))

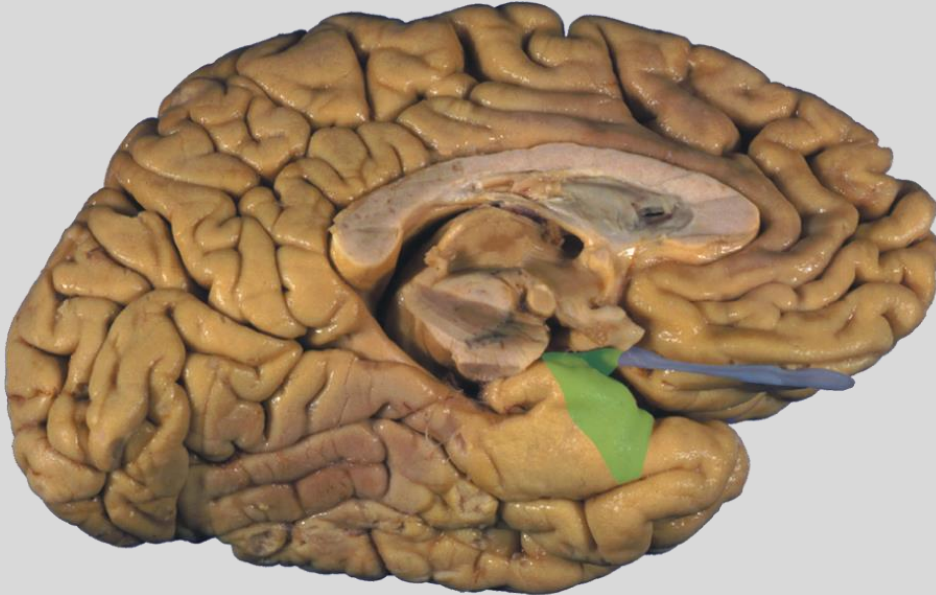
He could remember events and things that were consolidated years prior to the surgery.

What else did he lack?

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- He doesn't have normal **olfactory** threshold.....he is **unable** to **discriminate** among different **odors** or identify anything by smell – couldn't remember smells.
 - **Piriform cortex** (including the **entorhinal cortex** and **amygdala** mediates the perception of different odors and related emotional responses.
- He is impaired in **keeping track of his internal states**, presumably because of the removal of both **amygdala**.

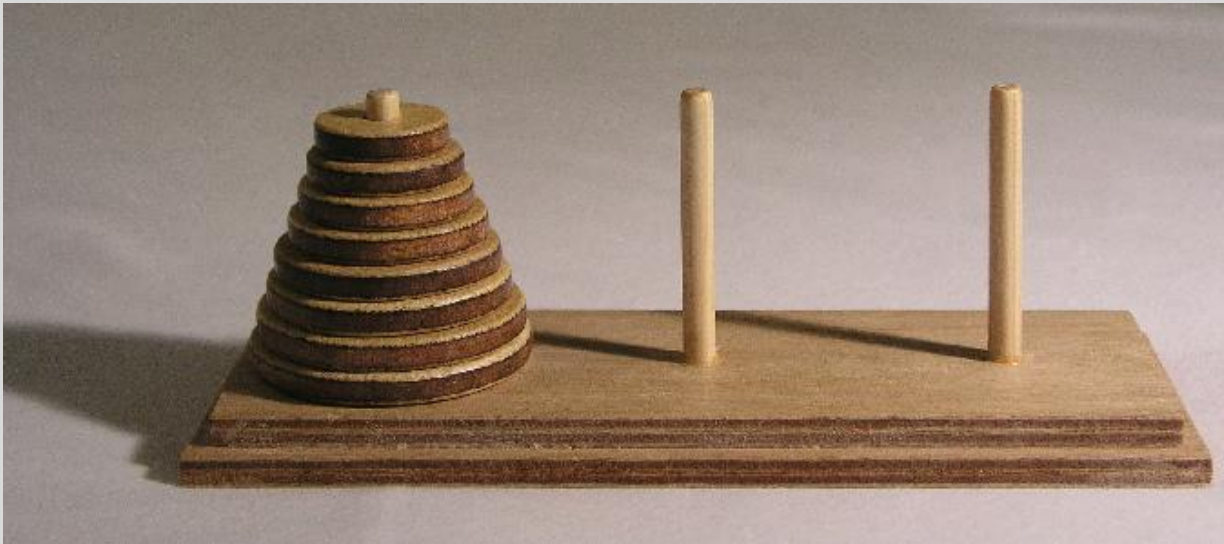


Tower of Hanoi

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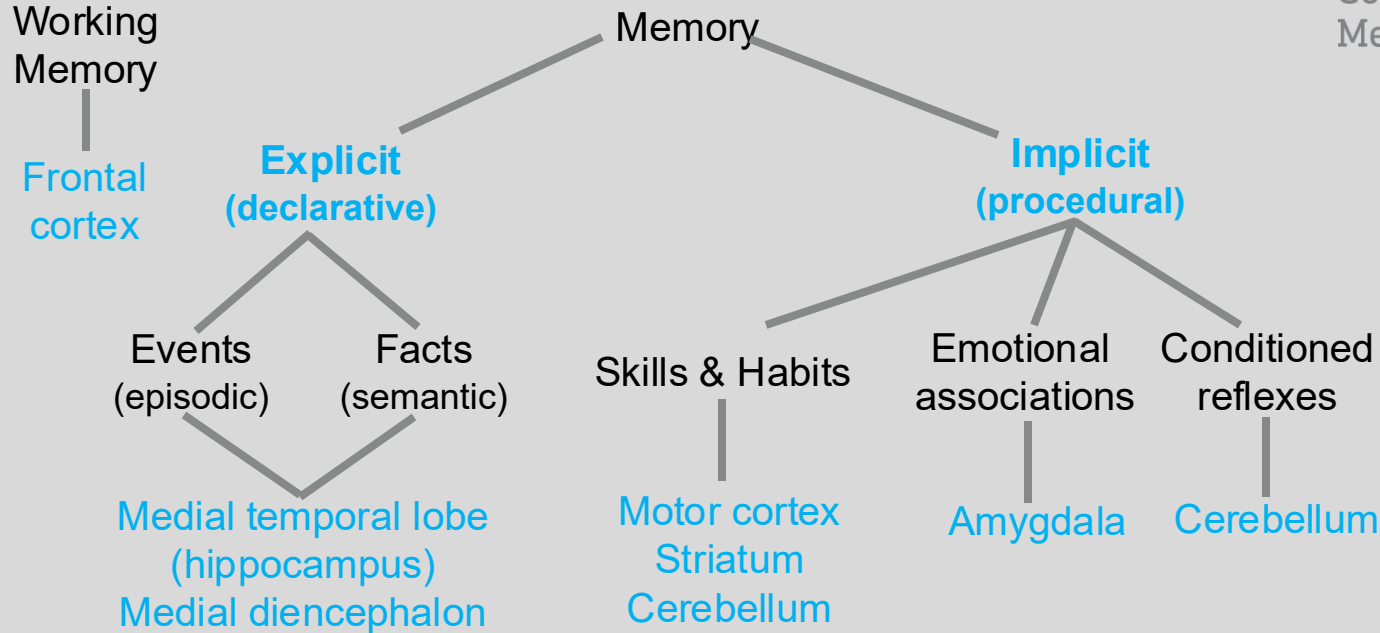
H.M. could remember how to perform puzzles
(Procedural Memory was good)



Different parts of the brain are responsible for different types of memory

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Long-term memory storage is thought to be in
diffuse areas of the Association Cortices

Memory Disorders

Amnesia

Alzheimer's Disease

Some Clinical Causes of Amnesia

Causes	Site of Damage
Vascular (PCAs)	Bilateral Medial Temporal Lobe (more sensitive to hypoxia)
Trauma (concussion, cerebral contusion)	Mammillary bodies, Mammillothalamic tract, Medial Thalamus
Surgery (for temporal lobe epilepsy)	Bilateral Medial Temporal Lobe
Infections (herpes simplex virus)	Bilateral Medial Temporal Lobe
Vitamin B1 Deficiency (Wernicke-Korsakoff's syndrome)	Mammillary bodies, Medial Thalamus

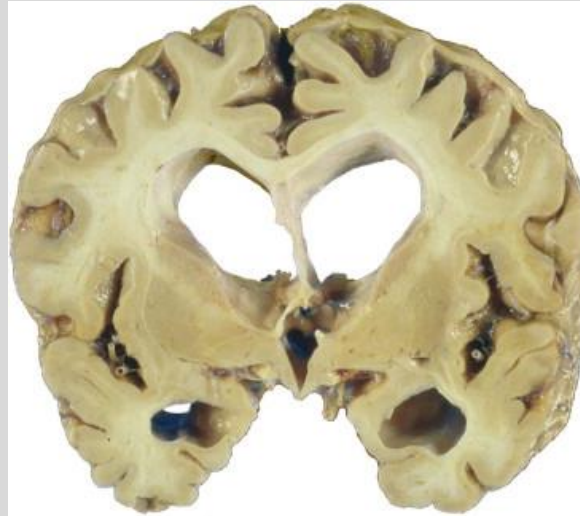
In all cases, the patients have bilateral injury to some component of their medial temporal lobe or thalamus.

Alzheimer's Disease

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- **Gray matter atrophy**, **hippocampal atrophy** and **ventricle enlargement**
- Significant **decrease in ACh**
- Significant degeneration of basal nucleus
- ACh antagonist produce deficiencies in memory tasks reversed by AChE Inhibitor
- **AChE Inhibitor** improves memory and attention in patients with Alzheimer's
- AChE Inhibitor also has been shown to decrease hippocampal atrophy



Test yourself

Question

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Match the type of memory with the CNS structure
(more than one structure may be matched to the same memory type)

Skills and Habits

Basal Ganglia

Working Memory

Amygdala

Emotional assoc.

Diencephalon

Conditioned
reflexes

Cerebellum

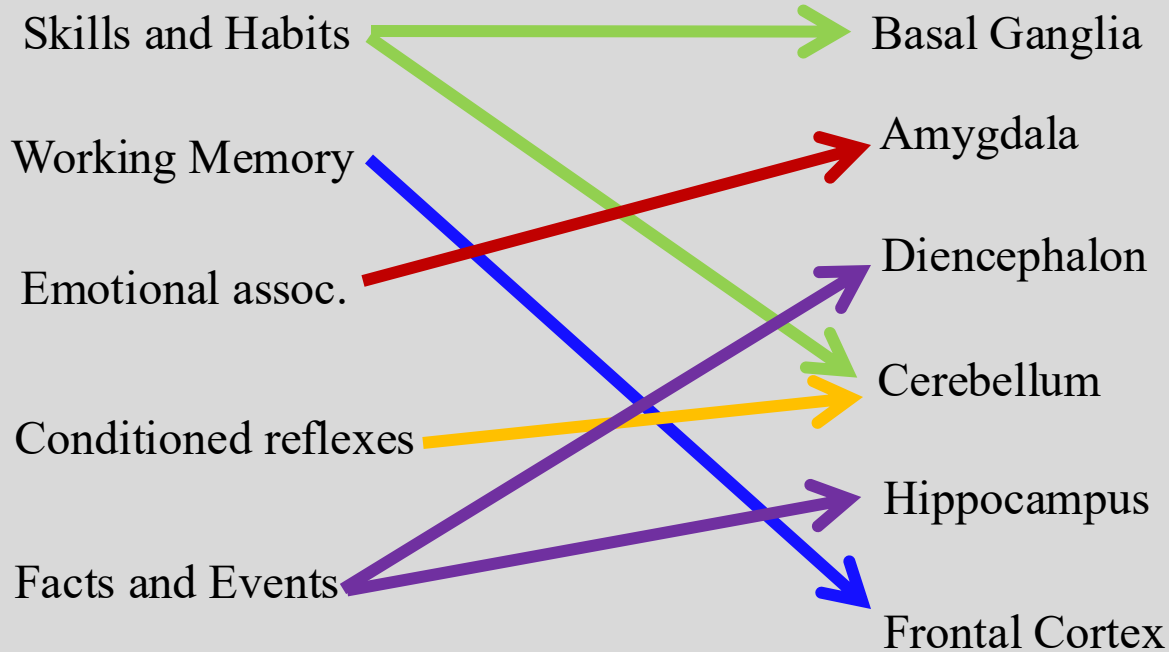
Facts and Events

Hippocampus

Frontal Cortex

Question

Match the type of memory with the CNS structure



Test yourself

Summary

Limbic Structure	Major Functions
Cingulate cortex	
Amygdala	
Hippocampus	
Septal area (Nucleus accumbens)	
Orbital frontal cortex	
Hypothalamus	

Summary

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Limbic Structure	Major Functions
Cingulate cortex	Depression, anxiety, pain
Amygdala	Fear, olfaction
Hippocampus	Memory, learning
Septal area (Nucleus accumbens)	Reward
Orbital frontal cortex	Secondary olfactory cortex
Hypothalamus	Homeostasis, autonomic regulation

Lecture Feedback Form _ Dr. Jennifer Xie

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



Email: jennifer.xie@nyit.edu